

Gustabo Lazaro

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Engineering Portfolio: https://glazaro3251-dev.github.io/Gustabo_Lazaro_ME/

EDUCATION

California State Polytechnic University, Pomona (CPP)

Expected August 2026

Bachelor of Science, Mechanical Engineering

Related Coursework: Manufacturing Processes, Mechanical Design, Engineering Materials, Measurements and Instrumentation, Finite Element Analysis (FEA), Control of Mechanical Systems, Engineering Analysis, Thermodynamics, Fluid Mechanics

WORK EXPERIENCE

United Rentals Inc. - Fontana, CA

June 2011 — Present

Service Technician III

- Inspect, troubleshoot, and repair mechanical, electrical, and hydraulic equipment/ tools to keep units safe, reliable, and ready for use.
- Use service manuals, schematics, inspection results, and equipment symptoms to identify problems and determine the proper repair approach.
- Document repair findings, parts used, completed work, and service recommendations to support accurate records and future troubleshooting.
- Work with service team members, parts personnel, management, and customers to prioritize repairs, resolve equipment issues, and return units to service.

STUDENT INVOLVEMENT

CPP NGCP (Northrop Grumman Collaboration Project) - Pomona, CA

August 2023 — Present

Payload Deployment Member

May 2025 — Present

- Overseeing cross-functional coordination with subsystems leads to track progress, resolving technical challenges, and maintain project schedules.

Payload Integration Lead Engineer

May 2024 — May 2025

- Led a team in the design, prototyping, and testing of UAV payload systems for aerial deployment.
- Built a MATLAB model for 3D parachute trajectory simulation to enhance drop accuracy and reliability.
- Designed and implemented a data logging system using Arduino, accelerometer, and SD card module to monitor and verify payload acceleration below 2Gs during deployment.

Payload Integration Member

August 2023 — May 2024

- Developed and tested autonomous payload release mechanisms in SolidWorks for a UAV supporting relief package delivery missions.
- Designed a payload enclosure with a parachute detachment mechanism by collaborating with Unmanned ground vehicles (UGV) to improve efficiency of picking up the medical relief package.
- Tested various parachute folding configurations and analyzed impact forces using energy-based methods; validated results with MATLAB Monte Carlo simulations.
- Conducted FEA and CFD analysis on the payload system to assess structural integrity and aerodynamic behavior during descent.
- Deployed the payload within 250ft of the specified release envelope, meeting overall project goals.

Finite Element Analysis Project — FEMAP/Nastran

- Prepared a simplified finite element model by removing small non-structural components and organizing major parts for analysis.
- Assigned material properties, shell thicknesses, mesh types, and joint connections to support structural analysis setup.
- Documented modeling assumptions, constraints, and analysis limitations as part of an individual and team-based FEA project.

Skills

CAD & Design: SolidWorks, AutoCAD 2D

Simulation & Analysis: MATLAB/Simulink, NX NASTRAN (FEMAP), SolidWorks Simulation

Other: 3D printing, Arduino IDE, Microsoft Office (Excel, Word, PowerPoint), Hands-On experience with hand tools and power tools, MIG Welding